

Prompt Surfaces in the Branch-Loaded Multimodal Skill Agent

The main agent decides whether to consult a skill; the temporary branch first gates visual evidence, then returns structured guidance.

temporary branch

no GUI action

Main Visual Agent Prompt

System prompt excerpt

Task skills are optional procedural guidance modules.

- Use skill hints before calling **LOAD_SKILL(...)**.
- **LOAD_SKILL** opens a temporary guidance branch.
- Branch does not execute the action for you.
- Skill hints and guidance notes are references only.
- Exhausted skills are removed from the list.

Output exactly one code block:

`pyautogui` code | WAIT | DONE | FAIL | **LOAD_SKILL(...)**

Per-step user prompt excerpt

Please decide the next grounded response for the **CURRENT screenshot**.

Instruction: {instruction}

Available skills:

name + description + minimal state hints

Active guidance:

skill_applicability, **plan**, **do_not_do**, **expected_state**

Use returned guidance before consulting again.

LOAD_SKILL

Branch Stage 1 Prompt

Gated view-selection policy

Stage 1: gated state-view selection.

Decide whether visual reference images are needed.
If yes, choose the evidence goal and minimal **views**.

Evidence goals:

locate_control -> one of full_frame/focus_crop
recognize_before -> before (+ optional full_frame)
verify_after -> after (+ optional full_frame)
compare_transition -> minimal non-default **views**

Return **LOAD_STATE_VIEWS**({...}) only.

Required branch output

```
LOAD_STATE_VIEWS({
  "visual_reference_needed": true,
  "why_not_text_only": "...",
  "requests": [{
    "state_id": "pyramid_shape_in_chart_type",
    "views": ["focus_crop"],
    "evidence_goal": "locate_control",
    "reason": "local control evidence is needed"
  }]
})
```

selected views

Branch Stage 2 Prompt

Guidance-only policy

Stage 2: guidance-only branch.

Do NOT return GUI actions.
Use **CURRENT screenshot** first.
Skill text, state bundles, Stage-1 decision, and selected **views** are references only.

Return **structured guidance** JSON only:

skill_applicability, **subgoal**, **plan**,
do_not_do, **fallback_if_no_progress**,
expected_state, **completion_scope**

Required guidance JSON

```
{
  "skill_applicability": "effective",
  "subgoal": "choose the requested chart subtype",
  "plan": "Use the live Chart Type page; select the structural subtype, then verify the visible chart before completion.",
  "do_not_do": "Do not copy reference coordinates.",
  "fallback_if_no_progress": "If the expected control is absent, backtrack to the chart wizard page.",
  "expected_state": "The requested control is visible.",
  "completion_scope": "needs_verification"
}
```

structured guidance returns to main context